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Effects of Carbon Nanotube Aspect Ratio on Strengthening and Tribological Behaviour of Ultra High Molecular Weight Polyethylene Composite

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Abstract

Carbon-based nanomaterials are great choice as reinforcement to ultra-high molecular-weight polyethylene (UHMWPE), with potential use in orthopedics. While high in-plane-stiffness and strength of these nanomaterials help in toughening, their weaker out-of-plane integrity offers lubrication. Present study investigates effect of aspect ratio of carbon nanotubes (CNT) on toughening and solid-lubrication efficiency of UHMWPE-matrix. A nominal 0.05–0.1wt.% of CNT addition increases hardness and elastic modulus of UHMWPE by 3-45% and 8-42%, respectively. Higher aspect ratio (HAR) CNTs are found more effective in improving hardness and modulus of UHMWPE. Wear rate and friction-coefficient also increase by 530% and 220%, respectively, while reinforced with HAR CNTs. Thermal analysis shows slight increase in crystallinity and stability of composite. HAR CNTs improve interfacial bonding with matrix, due to their morphological similarity to polymer chains, as compared to low aspect ratio CNT. Aspect ratio of CNTs significantly dominates strengthening and tribological behavior of UHMWPE.

Keywords: A. Polymer- matrix composites (PMCs), B. Wear, B. Mechanical properties, D. Thermal analysis.

1. INTRODUCTION

Ultra-High Molecular Weight Polyethylene (UHMWPE) has been extensively used in various joint replacement applications (artificial hip cups and tibial inserts), with articulating counteract surfaces made of metals or ceramics [1, 2]. UHMWPE has been chosen over other commercial polymers because of its superior toughness, impact strength, low friction and abrasive sliding wear resistance. In addition, easy fabrication ability and superior biocompatibility of UHMWPE in the physiological environment makes it even suitable for the above mentioned application [3]. However, tribological behavior of UHMWPE lining plays a critical role in deciding life span of modular acetabular cup in total hip implants [4]. Though UHMWPE is in clinical application for long time, its performance in extended period has been limited due to its inherent low yield strength, leading to permanent deformation, severe wear and ultimately premature failure of implant. Relative motion between the metal/ceramic

femoral head and polyethylene lining surfaces generates wear debris. Generated debris migrates to the tissues around the implant through blood and invites inflammatory reactions, leading to osteolysis and gradual aseptic loosening of the implant [5]. Thus, a drive always exists to improve the stiffness and wear resistance of UHMWPE lining without increasing the coefficient of friction (COF), which is a challenging goal in itself. Over the past decades, significant efforts have been taken up for the enhancement of tribological and mechanical properties of UHMWPE bearing. Various techniques are explored for this purpose, namely, cross linking of polymer [6-8], increasing the degree of crystallinity through ionizing radiation [9] and chemical methods [10]. These modifications decreased the risk of wear rate of UHMWPE. But a negative effect on ductility and strength is noted, resulting into catastrophic failure of the implant [11, 12]. However, main aim with the UHMWPE surfaces to be used in orthopedic joints is to have a unique combination of high wear resistance and low coefficient of friction, which apparently appears contradictory.

Polymer matrix composite is an alternate solution, where UHMWPE matrix is reinforced with either UHMWPE fibers itself [13] or various other carbonaceous nanoparticles and fibers, e.g., nanodiamonds, graphite and carbon nanotubes [3, 14-21]. Different processing routes, namely, solvent casting, ultrasonication, *in situ* polymerization and ball milling, are employed to synthesize these composites with an aim of improving mechanical and wear behavior of UHMWPE matrix [16]. Reinforcing with 5 wt. % of UHMWPE fibers increases the tensile strength (96 MPa) and elastic modulus (1.4 GPa) of the polymer matrix by three times and two times, respectively. Self reinforced UHMWPE composites have shown superior mechanical properties, but wear properties are found to be similar to un-reinforced UHMWPE [13]. Addition of 3 wt% graphene oxide to UHMWPE shows 44% increase in hardness and 70% decrease in wear rate with respect pure UHMWPE [22]. Whereas, addition of 1 wt. % graphene into UHMWPE, has shown 125% increase in elastic modulus and 100% increase in yield stress [14]. On the other hand, a very small amount (0.2 wt. %) of single wall carbon nanotube (SWCNT) addition to UHMWPE increases hardness and elastic modulus by ~ 66% and 58%, respectively [23]. Wear durability is also improved impressively in this composite by more than two orders of magnitude [23]. Hence, among all the reinforcement, carbon nanotube (CNTs) is a promising choice for polymer composite due to their exceptional properties, like, high elastic modulus (200–1000 GPa) and tensile strength (50–200 GPa) [24, 25], which helps in improving the mechanical behavior of polyethylene with minimum addition. As reinforcement to polymer, CNTs can also offer the unique combination of improved toughness and decreased co-efficient of friction on surface by acting as a solid lubricant. However,

effective utilization of CNTs in composite is strongly dependent on their uniform dispersion in the matrix, without disturbing their structural integrity [26]. Homogeneous reinforcement of phases and good interfacial bonding enhance the load transfer efficiency from matrix to CNTs and show improved mechanical properties in the composite [27]. Strong interfacial bonding between polymer matrix and CNT is highly influenced by the aspect ratio of CNTs [28]. Hence, aspect ratio of the reinforcement is considered as a dominating factor for developing the theoretical and experimental models [29]. A few studies have explored the influence of CNT diameter and aspect ratio on mechanical properties of polymer composites. Diameter of CNTs have shown significant effect on fatigue behavior of an epoxy – 2.5 wt.% multi wall carbon nanotube (MWCNT) composite. A 10 fold reduction in the fatigue crack growth rate is observed for CNTs having diameter in the range of 5-8 nm, while only 2 fold improvement was recorded for thicker CNTs with diameter of 20-30 nm [30]. Both the CNTs possessed similar length of 15 – 10 μm . Thus, fatigue resistance and fracture toughness of epoxy is found to be functions of the aspect ratio of the filler phase. Ayatollahi et al. [28] have studied the epoxy-0.5 wt% MWCNT composite with different aspect ratio of CNTs. A maximum of 32% increase in fracture toughness was observed with MWCNTs having aspect ratio of 1000 and it is decreased to 21% for a lower aspect ratio of 455. Both modulus of elasticity and tensile strength were increased with decreasing tube diameter. Small diameter and higher surface to volume ratio of CNTs form a percolated network in the composites and build strong interfacial bond with polymer chains, thus increasing the load transfer sites between the matrix and reinforcement [14].

Tribological properties of CNT reinforced UHMWPE composite is a highly relevant characteristic for potential bearing application. Only a few studies have dealt so far with the influence of CNT concentrations on wear and thermal stability of the polymer composite [31-37]. All of these studies explored effect of CNT addition on the surface hardness, strength, plowing and cutting resistance of UHMWPE matrix. The reports mostly include improvement in wear resistance and increase in coefficient of friction (COF). Few exceptions to this trend were also observed by Wei et al. [36] and Lee et al. [37], who reported decrease in COF by ~20 % with addition of 5 wt % CNTs. This decrease in COF is attributed to the self lubrication offered by CNTs. However, effect of aspect ratio of CNT in controlling the lubrication on polymer matrix surface is not reported.

As mentioned earlier, not many researchers have explored the morphological effect of CNT on mechanical properties of the composite. However, to the best of the authors' knowledge, no studies till date have dealt with the effect of CNT morphology on wear behavior of any polymer based composites. A thorough understanding of the

evolution in wear mechanism in presence of CNTs with different morphology, concentration and distribution is required to comprehend the role played by morphology of reinforcement.

In order to experimentally analyze the effect of CNT morphology on mechanical and tribological behavior of UHMWPE matrix, we used two different aspect ratios of MWCNTs, with two different concentrations, as reinforcement. Ball on disc tribometer and nanoindentation studies were conducted to study necessary properties required for artificial acetabular cup, e.g., COF, specific wear rate, hardness and elastic modulus. Differential scanning calorimetry and thermo gravimetric analysis was also carried out to study the thermal, oxidation or degradation behaviors of the composite structure. Analysis of the results gave us an in-sight on the origin of structural change occurring in composites, due to different aspect ratio of CNTs used as reinforcement. Finally, the obtained results were thoroughly analysed to understand the mechanisms dominating mechanical and tribological behavior of the composite.

2. EXPERIMENTAL PROCEDURE

2.1 Materials and Processing

Medical grade ultra high molecular weight polyethylene (UHMWPE) powder was procured from J.P polymer, Mumbai, India. It was in the form of white powder with an average granular particles size of 120 μm and density of 0.953 g/cm^3 . Two different aspect ratios of MWCNTs with more than 95% purity are used in this study. High aspect ratio CNTs (HARC), with outer diameter of 10-12 nm, length of 8-12 μm and density of 1.8 g/cm^3 were procured from CNano Technology Limited, Beijing, China. Low aspect ratio CNTs (LARC), with outer diameter of 40-70 nm, length of 1-3 μm and density of 2.1 g/cm^3 were procured from Inframat Corporation (Willington, CT). Their average aspect ratios (ratio of average length to diameter) are about 900 and 75, respectively for HARC and LARCs. Fig. 1a, b and c show FESEM images of as received CNTs as well as UHMWPE powder. An inset in Fig. 1a and b clearly reveals the difference in aspect ratios of two types of CNTs used in this study. Initially, CNTs were homogeneously dispersed in acetone using a probe sonicator for 60 minutes (power - 500 Watts; frequency - 20 kHz) followed by another 30 minutes sonication after mixing the UHMWPE powder to dispersed CNT solution. In order to avoid the agglomeration of CNT, polymer powder was added in dispersed CNTs slowly, while keeping ultrasonication on. In this process, CNTs, due to their high surface energy, gets attached to the polymer powder surface and while settling down they get uniformly distributed. The composite powders were dried completely in

keeping hot air oven at 60° C. Fig.1d and e show homogeneous dispersion and attachment of CNTs to UHMWPE powder.

Compositions with low reinforcement content (0.05 and 0.1 wt. %) were chosen for this study to avoid agglomeration. Following nomenclatures (given in parentheses) would be used for these different compositions throughout the manuscript: pure UHMWPE (PE), UHMWPE-0.05 wt. % of HARC (PE-0.05HARC) and LARC (PE-0.05 LARC), UHMWPE-0.1 wt. % of HARC (PE-0.1HARC) and LARC (PE-0.1LARC). Disk shaped pellets of 20 mm diameter and 5 mm thickness were fabricated by compression molding of the composite powder at a pressure of 300 MPa at room temperature, followed by curing at 160° C for 60 minutes in hot air oven.

2.2. Characterization

The literatures highlighted that the crystallinity and degradation behavior relatively influences the thermal and mechanical property of polymer based composites. Hence differential scanning calorimetry (DSC) and thermogravimetric analysis (TGA) was performed using SEIKO 6300 EXTAR TG/DSC instrument. About 10 mg of the test samples was sealed in aluminum pans, followed by heating from 35 to 500° C at a heating rate of 5° C/min under nitrogen atmosphere. Degree of crystallinity was calculated using the following equation [38]:

$$X_C = \frac{\Delta H_C}{\Delta H_{100}} \times 100\% \quad \text{..... (1)}$$

where, X_C is degree of crystallization, ΔH_C is the heat energy per unit mass of the composite. Enthalpy of fusion, ΔH_{100} , is found to be 289 J/g for a 100 % crystalline UHMWPE [39].

In order to understand the effect of CNTs distribution in the composite matrix and within the wear tracks, field emission scanning electron microscope (FE-SEM, Carl Zeiss Ultra Pluss, Germany) was used.

2.3 Mechanical and Tribological Characterization

Nanoindentation test was conducted to measure the mechanical properties of the composite using Hysitron TI950 Triboindenter (Hysitron Inc, USA) equipped with three-sided Berkovich diamond indenter with a tip radius of 100 nm. Indentations were made in the depth controlled mode to achieve same depth of penetration in all the indents. The maximum penetration depth of 150 nm was achieved in 10s followed by holding for 2s after attaining the peak displacement. A minimum of 25 indents were taken on each pellet to evaluate the hardness (H) and reduced elastic

modulus (E_r). E_r is calculated from the slope of the initial portion of unloading curve of load versus displacement plots using Oliver and Pharr method [40].

Ball-on-disc tribometer (TR-201E-M2, DUCOM, Bangalore, India) test was performed to evaluate the friction and wear behaviour of the UHMWPE-CNT composite. One of the intended applications of this UHMWPE based material system is in acetabular cup lining of hip implant. This lining goes through several rubbing action with the femoral head, which can be properly addressed through the macro scale wear studies with the rounded (spherical) surface. That is why ball-on-disc tests were chosen for evaluating tribological behavior of the composite. Commercially available EN-31 forged steel ball (hardness - 63 HRC) with 6 mm diameter was used as counter body. Normal load and speed used for the tests were 5 N and 200 rpm, respectively. The track diameter was approximately 6 mm and test duration was 2 hour. At least three tests were performed in each composition. Contact profilometer (SJ 400, Mitutoyo, Japan) was used to determine the depth and width of the wear track. At least ten profiles across the track were recorded in each of the three tracks for all the compositions to measure the average width and depth of the wear track. Wear volume (V) of the track was calculated using the following formula.

$$V = 2\pi \cdot r \times A \dots\dots\dots (2)$$

where r is the radius of the wear track; A is the cross sectional area of the wear track obtained by integrating the groove in the profile of the wear track. The variation in wear volume, thus measured, is reported as error bars in the corresponding calculations of wear rate.

Electronic sensor was attached to tribometer to record the continuous frictional force. The co-efficient of friction with respect to sliding distance is obtained from there. The specific wear rate was calculated from the following relationship;

$$W = \frac{V}{P \times S} \dots\dots\dots (3)$$

where W is the specific wear rate; V is the total wear volume; P is the normal load and S sliding distance travelled in 2 hours. The worn surfaces were thoroughly studied through FE-SEM to identify the mechanisms dominating their tribological behavior.

3. RESULTS AND DISCUSSION

3.1 Microstructural Characterization of the Composite

Mixing of initial powders using probe sonication (power-500 Watts and 20 kHz frequency) is found to be very effective in distributing the CNTs very uniformly with PE powders, irrespective of their significant difference in

sizes. Fig 1d and e show CNTs getting homogeneously distributed and attached to the polymer powder surface as a result of probe sonication. This is very important for effective use of the total reinforcement volume and uniform mechanical/tribological behavior of the composite. The curing temperature of the polymer and composites was determined through differential scanning calorimetry (DSC) of the as-received PE powder (Fig.2). The melting temperature of PE is nearly 123° C, but it is very viscous at this point. Whereas, proper curing and wrapping of the reinforcement phase by polymer matrix needs more flow and lesser viscosity. Thus a higher temperature of 160°C was used for curing. This was optimized by trial and error at a range of temperatures and by observing the integrity of cured structure through microscopic investigation at the cross section of the PE pellets. Those results are not included in this study, as they are not found very significant to the objective of this study other than just optimizing the curing temperature.

Fig. 3a and b show the fracture surface of the HARC and LARC reinforced composite with 0.1 wt. % CNT content. CNTs are nicely embedded with polymer matrix in both the composites. Both types of CNTs are found to be deeply rooted in the matrix with protruded ends revealed at fracture surface, which indicates strong interfacial bonding with the polymer matrix resulting in effective load transfer. However, the protruded ends are found to be longer in case of HARC than in LARCs. The reason could be the lower aspect ratio of the latter, and/or uprooting of the same at higher load, as compared to the former.

3.2 Thermal Analysis of the Composites

Fig.4 shows the DSC results of the composite. The peak in the DSC curve corresponds to the melting temperature and area under the peak is equal to the enthalpy of melting. Researchers [41-43] have reported that CNTs can act as nucleating agent for the crystallization of polymer. Addition of a mere 1 wt. % MWCNTs also enhance the rate of crystallization up to a significant distance from their vicinity. As a result, crystallinity of polymer increases without a change in melting temperature of the composite. On the contrary, reduction in crystallinity of polyethylene matrix from 58 to 48 % for 5 wt. % CNT [3] and 32.5 to 27.6 % for 10 wt. % of CNT [44] is also reported by other researchers. However, no study has reported yet the effect of aspect ratio of CNT, if any, on the crystallinity of polymer matrix composite. To understand this, DSC tests were conducted and the results were analyzed. The enthalpy of melting for pure PE, PE-0.05LARC and PE-0.1LARC is 88.1 J/g, 84.5 J/g and 70.3 J/g, respectively (Fig.4). The percentage of crystallinity was calculated by dividing the enthalpy of melting of the prepared composite

with the enthalpy of melting for 100 % crystalline PE [38], following equation 1. Crystallinity recorded is 30 %, 29 % and 25 %, respectively, for PE, PE-0.05LARC and PE-0.1LARC. Addition of lower aspect ratio CNTs reduces the crystallinity of the composite to some extent, which can be due to two reasons. Firstly, LARCs are dispersed uniformly in PE matrix as stiff fibers (Fig.1e), which might form tube to tube connection in some cases. This network enhances the thermal conductivity of the composite owing to the excellent thermal conductivity of 3,000 W/m·K [45] for CNTs. As a result a net increase in the cooling rate of the composite during solidification is attained, leaving less time for arrangement of the polymer chains to form crystallites. This type of behavior is also observed by other researchers [3, 46]. Secondly, LARCs have higher diameter as compared to polymer chains and hence they act as obstacles and restrict the movement of chains during solidification. This may be other reason for reduction in the crystallinity of the LARC reinforced composites. Agglomeration of nanotubes could have been another reason for decrease in crystallinity. However, significant agglomeration of nanotubes was not observed in the fracture surface of the composites (Fig. 3) in the present study.

PE-0.05HARCs show almost similar crystallinity of 29.76% as PE. However, high concentration of HARCs (PE-0.1 % HARC) shows increase in crystallinity by ~ 3 %. One of the reasons for the latter case could be that randomly bent fibers and self entangled flocks of high aspect ratio CNTs cover the polymer. Further, their higher surface area to volume ratio (as compared to LARCs) increases the viscosity and nucleation sites for the crystallization in the composite and hence crystallinity improves. Further, HARCs are nicely embedded within matrix due to their morphological similarity with polymer chain and it is discussed in detail later in subsection 3.3. The higher specific heat of CNTs ($C_p=0.75 \text{ Jg}^{-1}\text{K}^{-1}$ at 300 K [47]) can cause a heat pool in the surrounding polymer, which thermodynamically helps rearranging the polymer chains in a manner to increase the crystallinity. But 0.05 wt. % is not a sufficient filler fraction for giving such effect and hence PE-0.05HARC has shown almost no change in crystallinity. This observation is in line with the findings of other researchers [43, 48]. Therefore, aspect ratio and volume fraction of the CNTs influences the crystallinity behavior of the composite, which indirectly affects the mechanical properties.

Effect of CNT aspect ratio on thermal degradation behavior of the composites can be explained from the TGA plots presented in Fig.5. It is observed that high aspect ratio carbon nanotubes reduce the amount of oxidation or degradation of the composite at higher temperature. At 400° C, the mass remaining for PE, PE-0.05HARC and PE-0.1HARC is 81.13 %, 90.91 % and 92.45 % respectively. This is due to high aspect ratio and surface area of CNTs

which are sufficient to cover the polymer particles and to give protection from degradation. Further, more carbon – carbon bonds in HARC and their self entanglement do not allow the polymer to melt. In addition, HARC reinforcement also induces more crystallinity to the polymer matrix, as discussed before, which makes the structure more stable at higher temperature. As a result, HARC reinforced composite shows more stability at higher temperature as compared to pure polymer. In case of LARCs, the mass loss is much higher in composite than pure PE, with a 79 % remaining for PE-0.1LARC and 71 % for PE-0.05LARC at 400° C. It is due to the fact that uniformly dispersed stiff fibers expose their surface freely resulting in easy oxidation of amorphous carbon present in the CNTs and other carbonaceous impurity. Since the LARCs have much mismatch of size with that of polymer chains, there is a high chance their surfaces will be exposed and oxidized more than the HARC, as the latter would be wrapped more by polymers. Further, LARC reinforcement decreases the crystallinity by disturbing the polymer chain arrangement, which leads to increase in amorphous region. Polymer chains in amorphous regions easily degrade when it is exposed to higher temperature. Hence, more mass loss is found in composite with less concentration of LARCs as compared to pure PE. But, increasing the concentration of LARCs reduces the mass loss to some extent, because more CNTs will stabilize the composite by covering polymer and also with increase in CNT content, the population of C-C bonds increases and as these bonds does not degrade easily at the operating temperature. These two becomes competing phenomenon, which causes some decrease in degradation rate at PE-0.1LARC as compared to PE-0.05LARC.

As a whole, it can be conceived that even low concentration of HARC are also capable of significantly improving the thermal stability of polyethylene matrix.

3.3 Effect of CNT Aspect Ratio on Hardness and Elastic Modulus

The mechanical properties of the composites are evaluated through instrumented indentation technique. Fig.6 shows the representative load-displacement plots obtained from nano indentation test of different compositions with a depth controlled indentation mode. It can be observed that the load required to attain 150 nm depth of indent is higher for all the CNT reinforced PE composite. Presence of CNTs in polymer matrix helps in sharing load and thus the composites show higher hardness and elastic modulus. Fig.7 shows the average hardness and elastic modulus values and their standard deviations for at least 25 indents on each composition. Improvement in hardness and modulus is observed in the range of 3-45% and 8-42%, respectively, with increasing the concentration of different

aspect ratio CNT reinforcement. PE-0.05HARC shows 13% and 29 % improvement in hardness and elastic modulus as compared to pure PE. But, in case of PE-0.05LARC, a smaller improvement of 3% in hardness and 8% in elastic modulus is recorded. These results clearly indicate that HARC has an edge over LARC in terms of mechanical behavior of the composite. Same trend is observed in case of composite with 0.1 wt. % nanofillers. PE-0.1HARC shows an impressive improvement of 45% in hardness and 42% in modulus, whereas the improvement in case of PE-0.1LARC is 19% in hardness and 28% in elastic modulus over pure PE. Appreciable improvement in mechanical properties for composites reinforced with HARC can be explained in terms of their high surface area to volume ratio and interfacial strength. These factors increase the load transfer efficiency to the CNT network throughout the composite uniformly and avoid the failure due to high stress concentration. Cipiriano et al. [48] have also noticed positive effect of aspect ratio of CNT on storage modulus, elastic modulus and viscoelastic property of polystyrene/CNT composite. HARC, in the present study, has diameter 10-12 nm, which is in much closer range to typical polymer chains with diameter range of ~1-3 nm [49]. This opens up two possibilities. Firstly, the HARC can align themselves along polymer chains, increasing a better interfacial bond area and thus more effective load transfer. Secondly, due to the similar diameter and the broken C-C bonds present in CNT, they can act as cross linkers between polymer chains and making the inter-chain sliding difficult. This in turn restricts the plastic deformation and increases the strength of the polymer. In case of LARC, due to their huge mismatch in size (diameter) with the polymer, they cannot align or act as cross-linkers. Rather, they are potential cause of disturbance in the arrangement of polymer chains, decreasing the strength. This hypothesis is explained with the help of a schematic presented in Fig.8.

Further, at identical composition level, the high aspect ratio CNTs have more number of total floccs as compared to low aspect ratio CNTs, resulting in percolation network with compact structure. This increases the micro mechanical interlocking of CNT and polymer chains, holding the matrix together and restricting the movement of polymer chains through bridging between chains. In case of LARC, chances of interfacial bonding and mechanical locking are lesser due to their larger diameter and shorter length. Thus, the mechanisms of strengthening in HARC are manifold, as compared to LARC with only sharing of load through interface. This explains the significantly higher hardness and elastic modulus in polymer matrix with HARC reinforcement as compared to LARC. The reinforcement potential of fibers, as a function of their aspect ratio and volume fraction, has been explained by some theoretical models. Amongst these, shear lag based models and Halpin-Tsai model are widely used. A modified

form of Halpin Tsai relationship, as proposed by Kanagarj et al. [43] and Coleman et al. [50], gives a simpler approach getting rid of information related to the thickness of interfacial region and fiber alignment direction. Thus, this one is more suitable for present study with randomly oriented nanotubes well adhered in the polymer matrix. In this model, both volume fraction and aspect ratio of CNT are considered to estimate the elastic modulus, as following:

$$E_C = (\eta_1 \eta_o E_{CNT} - E_{PE}) V_{CNT} + E_{PE} \quad (4)$$

$$\text{where, } \eta_1 = \text{length efficiency factor} = 1 - \frac{\tanh(a - 1/d)}{(a - 1/d)}, \quad a = \sqrt{\frac{-3E_{PE}}{2E_{CNT} \ln(V_{CNT})}}$$

η_o = Orientation efficiency factor = 0.2 for randomly oriented fiber; Young's modulus of CNT is 910 GPa [51] and for PE is 0.95 GPa (from the present study).

The above expression is used for estimating the theoretical elastic modulus of different compositions used in this study. Fig .9 shows comparison of experimental and theoretical elastic modulus for the two selected aspect ratios of CNTs. It can be observed that modified Halpin-Tsai model predicted values are in the similar range to that of the experimental values. Further, regression analysis is done to verify how this experimental result fits to the theoretical results. R^2 (coefficient of determination) values are calculated for both HARC and LARC in different volume fraction. The R^2 value for HARC and LARC is 0.822 and 0.957, respectively, which indicates linear relationship between the predicted theoretical and experimental results of the composite. Halpin Tsai equation fits very well for low volume fraction. But linear increase of modulus is predicted for higher volume fraction [50] as shown in fig.9. Thus, modified Halpin-Tsai model is a better fit for this study, as the volume fraction of reinforcement is < 0.1 wt. % in all cases. High aspect ratio CNTs have shown high modulus because a good fraction of isolated nanotubes are potentially aligned with polymer chains, which leads to high interfacial bond and cross linking of polymer chains. Fig. 8 clearly shows HARC aligned along the direction of polymer chains in fracture surface of the composite. The strong coating of polymer around the CNT and their interfacial bonding can easily transfer the load on matrix to CNTs and help in avoiding the rupturing of polymer chains as well as in minimizing the stress concentration in the composite.

3.4 Effect of Aspect Ratio of CNT on Tribological Properties

Fig.10 shows the change in co-efficient of friction (COF) as a function of sliding distance at different compositions of UHMWPE-CNT composite. The COF increases from zero to peak value during running-in-period due to the influence of surface roughness and thereafter attains a steady state. It is observed that addition of CNTs to polymer remarkably reduces the friction force as compared to pure PE composite. One of the key reasons is the role played by carbon nanotubes as solid lubricants, by releasing graphene layers on wear track [52]. MWCNTs possess concentric cylindrical layers of graphene, bonded to each other with Vander waals forces [53]. Individual cylinders of multi wall carbon nanotubes are expected to slide or rotate easily with respect to each other because MWCNT interlayer corrugation energy is small. Therefore, it is expected that decreased friction is due to worn off CNTs acting as a lubricants, like solid powder graphite lubricants [35, 52-54]. In addition, the open part of the reinforced CNTs, which are protruding and dangling on surface, can also act as lubricator due to their smooth cylindrical surface that gives a rolling action. Aspect ratio of the CNTs greatly influenced the friction force irrespective of the concentration. HARC has shown 120 and 25 % lesser COF as compared to LARC at 0.1 and 0.05 wt. % of concentration, respectively (Fig.10). The percolation network formation of CNTs strongly depends on the aspect ratio. It is expected that the HARC is rooted deeply and strongly in the polymer matrix and it contributes to continuous lubrication effects. Fracture surfaces of the composites have also shown longer HARC protruding out of matrix as compared to LARC (Fig. 3). This in turn signifies the presence of CNTs in wear track for longer time to be able to release more graphene layers. On the contrary, LARC tends to break from the polymer matrix and get detached from the track. Thus, they cannot provide as efficient lubrication as HARC.

The surface profiles of the wear tracks were measured in the transverse direction of the sliding. Fig.11 furnishes the representative profiles of wear tracks in each composition. The average width and depth of the PE track (2.5mm and 35 μ m respectively) are much higher than that of PE-0.05LARC (2mm and 30 μ m) and PE-0.05HARC (1.5mm and 17 μ m). The width and depth of the track on PE-0.1LARC is 1.5mm and 20 μ m, respectively as compared to those in PE-0.1HARC being 0.7mm and 4 μ m. Furthermore, the width and depth of the track decreased remarkably with increasing CNT concentration. All together, this shows significant improvement in wear resistance with CNT concentration, as well as increasing aspect ratio of CNTs.

The variation in specific wear rate with composition is shown in Fig 12. Addition of 0.05 wt. % of HARC and LARC reduces the wear rate by 76% and 18%, as compared to PE. Addition of 0.1 wt. % reinforcement reduces the

wear rate by 530% and 84%, respectively, for HARC and LARC. This enormous improvement in wear resistance is expected to be a combined effect of increase in mechanical properties and decrease in COF.

The worn surfaces of the prepared composites were examined to understand the predominant wear mechanism (Fig. 13). Continuous sliding causes increased polymer asperities at the tribocontact with the harder counter body. Further, due to rubbing action, the elastic plastic deformation of polymer asperity increases the deep furrows and it further rolls into thin molecular sheets. This results in deeper grooves and generates wear debris on the polymer surface. Deep craters and damages on the rough wear track of PE (Fig. 13a) indicate low wear resistance of this surface. Hard asperities and compaction of wear debris (Fig.13a') lead to increase in COF. But, in case of CNTs reinforced composite, smoother wear track surface without broad exfoliation and microcracking (Fig.13 b and c) is observed. Strong interfacial shear resistance of the CNT reinforced composite restricts the ploughing and cutting mechanism. The effective load transfer from polymer to CNTs suppresses the plastic deformation and cracking, which leads to less removal of the polymer particles from the composite. In addition, continuous lubrication effect from CNTs results into a smoother worn surface. CNTs get exposed on the surface of the composite during sliding, creating the interface and preventing direct contact between the counter body surfaces, which greatly reduce the ploughing and scratching effect in the composite. This effect is predominant in case of HARC, because these CNTs are deeply rooted within the matrix (Fig 13c'). On the other hand, LARCs come out with the polymer coating. Thus, they fail to give continuous lubrication effect and also increases the debris in the surface (Fig.13b').

While comparing the effect of CNT aspect ratio on wear behavior, HARC is found to show significant improvement over LARC at both the concentration. HARC composite (Fig 11) shows shallower and smoother wear track with less width and depth as compared to LARC composite. Smoother wear tracks indicate that the tribological loss is mainly governed by adhesive wear, which results in less wear volume. The effect is more prevalent with increasing nanofiller content, noting an impressive decrease in specific wear rate of 49% and 243% at 0.05 and 0.1 wt. % of HARC content, respectively, as compared to similar concentrations of LARCs (Fig. 12). The additional improvement in wear resistance for HARC reinforced composite (w.r.t. LARCs) can be explained in terms of better mechanical behavior and lower COF in the former. The strong interfacial bonding of HARC to polymer chains and their self entangled compact structure prevents the easy removal of debris and relative movement of polymer chains. All the above discussions point towards the benefit of using HARC as reinforcement in PE composite for improved tribological behavior.

4. CONCLUSIONS

In the present study, CNTs with two different aspect ratios (HAR-900 and LAR-75) were reinforced with PE matrix. Composite having higher aspect ratio of CNTs has shown greater hardness and elastic modulus as compared to CNTs with lower aspect ratio. Differential improvement in mechanical properties of composite with HARC and LARC is explained in terms of high surface area to volume ratio and strong interfacial bonding in the former, which increases the load transfer ability from matrix to CNT. Thermal stability of the composite containing HARC is also better as compared to LARC. Tribological behavior of the composites reveals significantly lower coefficient of friction and specific wear rate for HARC reinforced composites, which is the ideal requirement for its intended application in orthopedic joints. Better attachment of HARC to polymer matrix and their morphological similarity to polymer chains are found to be important factors in fortifying their dual role as toughener and solid lubricator for polymer matrix.

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Figure captions

Fig. 1 FE-SEM images of a) high aspect ratio CNT b) low aspect ratio CNT and c) UHMWPE (PE) powder. Figure (d and e) shows the HAR and LAR CNT dispersion on PE powder, respectively, before sintering.

Fig. 2 DSC curve recorded with PE powder

Fig. 3 Fracture surface of 0.1 wt. % HARC (a) and LARC (b) composite

Fig. 4 DSC analysis for different CNT reinforced PE and pure PE composite.

Fig. 5 Thermo gravimetric curves of composite revealing degradation loss through temperature range up to 500° C

Fig. 6 The representative load displacement curves obtained from nanoindentation tests for prepared composite

Fig. 7 Hardness and Elastic modulus of different compositions obtained from nanoindentation

Fig. 8 Graphical representation of the effect of CNT aspect ratio while interacting with polymer chains

Fig. 9 Regression analysis for the experimental and theoretical elastic modulus of the HARC composite (a) and LARC composite (b)

Fig. 10 Variation of friction co-efficient over the sliding distance for the different compositions

Fig. 11 Wear track surface profile for the prepared composites

Fig. 12 variation of Specific wear rate for prepared composites

Fig. 13 FE-SEM images of worn surface showing debris contained rough surface in pure PE (a and a') and smooth features in PE-0.1LARC (b-b') and PE-0.1HARC (c-c')

























